Chapter 1

Real Numbers

1.1 Introduction

The chapter Real Numbers deals with the set of numbers which are commonly used in our day-to-day life. It also tries to explain the different types of numbers and the properties that govern them.

Important Points

Real numbers include all rational and irrational numbers.

Rational numbers can be written in the form p/q, where p is an integer and q is a non-zero natural number.

Irrational numbers cannot be written in the form p/q.

There exist rational numbers which are not natural numbers, and natural numbers which are not rational numbers.

Examples: sqrt(2), pi, 3, and 0 are all real numbers.

1.2 Rational Numbers

A number which can be written in the form p/q, where p is an integer, q is a natural number and q is not equal to zero, is called a rational number. The set of rational numbers is denoted by the symbol Q.

Examples of rational numbers: 0.333..., 2/3, -7/5, 0, 5, -2, 0.2

Every integer is rational, because it can be written as p/1.

A rational number can be terminating or non-terminating in its decimal form.

1.3 Irrational Numbers

Another important class of numbers is the set of irrational numbers. If x is not a rational number, then it is irrational. The set of irrational numbers is usually denoted by I.

sqrt(2), sqrt(5), sqrt(3) are all irrational.

The decimal expansion of an irrational number is non-terminating and non-repeating.

There is no general way of knowing whether a given number is irrational.

1.6 Worked Examples

Example 1. Express 0.333... in the form p/q, where p is an integer and q is a natural number.

Let x = 0.333...

10x = 3.333...

Subtracting, 10x - x = 3.333... - 0.333...

Therefore 9x = 3 and x = 3/9 = 1/3.

Example 2. Prove that sqrt(5) is irrational.

Assume, to the contrary, that sqrt(5) is rational.

Then sqrt(5) = a/b, where a and b are coprime integers and b is not 0.

Squaring, 5 = a^2 / b^2, so a^2 = 5b^2.

This forces 5 to divide a^2, hence 5 divides a, so a = 5c.

Then 25c^2 = 5b^2, giving b^2 = 5c^2, so 5 divides b as well.

This contradicts the assumption that a and b are coprime.

Therefore sqrt(5) is irrational.

1.7 Practice Set

State whether each of the following is rational or irrational: 7/9, 0.121212..., sqrt(11), 2.6456756..., 0.

Find three rational and three irrational numbers between 1 and 2.

Prove that sqrt(3) is irrational.

Express 0.101010... in the form p/q.

Is every natural number a rational number? Justify your answer.

Show that 5 + 3sqrt(2) is irrational.

Important Formulas

p/q, where p is an integer and q is a non-zero natural number, defines a rational number.

A decimal that terminates is rational; a non-terminating repeating decimal is rational.

A non-terminating, non-repeating decimal is irrational.

n is a perfect square if n = m^2 for some integer m; then sqrt(n) is rational.

If n is not a perfect square, sqrt(n) is irrational.

The square root of a negative number is not real.

Key Definitions

Rational number Q: numbers of the form p/q.

Irrational number: any real number that is not rational.

Real number R: Q together with the irrationals.

Number line: an infinite line with a point for every real number.

Positive and negative: every non-zero real number is either greater than or less than zero.

Exam Tip

When a question asks you to prove a number is irrational, always begin with a contradiction and finish by showing two coprime integers share a factor.

NCERT Solutions

Step-by-step solutions to the exercise questions

Exercise 1.1 Question 1

Express each number in the form p/q, where p is an integer and q is a natural number and q is not equal to zero.

0.333... : Let x = 0.333..., then 10x = 3.333..., so 9x = 3 and x = 1/3.

0.272727... : Let x = 0.2727..., then 100x = 27.2727..., so 99x = 27 and x = 3/11.

0.454545... : Let x = 0.4545..., then 10x = 4.5454..., so 9x = 4 and x = 4/9.

0.818181... : Let x = 0.8181..., then 100x = 81.8181..., so 99x = 81 and x = 9/11.

Exercise 1.1 Question 2

Classify the following numbers as rational or irrational.

sqrt(2), sqrt(5), sqrt(11), sqrt(15), sqrt(21) are all irrational (none is a perfect square).

2/3, 7/2, 0.5, 0.444..., 0.2727... are all rational.

pi is irrational because its decimal expansion never terminates and never repeats.

~~Always state the substitution clearly. Marks are awarded for the setup, the linear equation and the~~ final simplification write all three.

Extra Practice Questions

With answer key

Section A One mark

Define a rational number.

Give one example of an irrational number.

Is 0 a rational number? Why?

Write three rational numbers between 0 and 1.

Section B Two marks

Express 0.666... in the form p/q.

Show that 2 + 5sqrt(3) is irrational.

Find five rational numbers between 2 and 3.

Find the smallest natural number n such that n + 1/2 is rational.

Section C Three marks

Prove that sqrt(6) is irrational.

Answer Key

0.666... = 2/3

2 + 5sqrt(3) is irrational

Rational examples between 2 and 3: 5/2, 11/4, 8/3, 21/8, 13/5

sqrt(6) is irrational (6 is not a perfect square)

0.370370... = 10/27